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A Scheduling Algorithm for Appliance Energy Consumption Optimization in a Dynamic Pricing Environment

Hamza Touhs ¹, Anas Temouden ¹, Ahmed Khallaayoun ¹ , Mhammed Chraibi ¹ and Hamza El Hafdaoui ^{1,2,*}

¹ School of Science and Engineering, Al Akhawayn University, Ifrane 53000, Morocco

² National School of Applied Sciences, Sidi Mohamed Ben Abdellah University, Fez 30000, Morocco

* Correspondence: h.elhafdaoui@au.ma

Abstract: This research delves into the intricate landscape of energy scheduling and optimization within microgrid and residential contexts, addressing pivotal aspects such as real-time scheduling systems, challenges in dynamic pricing, and an array of optimization strategies. This paper introduces a cutting-edge scheduling algorithm, harnessing the power of artificial neural networks driven by Long Short-Term Memory Networks, and highlights its exceptional performance, boasting a significantly lower Mean Absolute Error of 5.32 compared to conventional models. This heightened predictive accuracy translates into tangible improvements in both energy efficiency and cost savings. This study underscores the delicate balance between user satisfaction, cost reduction, and efficient scheduling for sustainable energy consumption, showcasing a remarkable 38% enhancement in optimized schedules. Further granularity revealed substantial gains in energy efficiency and cost reduction across different scheduling intensities: 11.11% in light schedules, 20.09% in medium schedules, and an impressive 38.85% in heavy schedules. However, this research does not shy away from highlighting challenges related to data quality, computational demands, and generalizability. Future research trajectories encompass the development of adaptive models tailored to diverse data qualities, enhancements in scalability for and adaptability to various microgrid configurations, the integration of real-time data, the accommodation of user preferences, the exploration of energy storage and renewables, and an imperative focus on enhancing algorithm transparency.



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1. Introduction

Drawing from a multitude of theories, the discernible advancement of real-time scheduling systems stands out, a field that has been particularly propelled by the progress achieved through fixed-priority scheduling theories. These theories have notably empowered the efficient prioritization of tasks under a spectrum of constraints. This scholarly work embarks on an exploration of the intricate landscape of scheduling challenges, placing particular emphasis on the complexities inherent in resource-constrained project scheduling. Specifically, it delves into the nuances of the multi-mode Resource-Constrained Project Scheduling Problem (RCPSP). Beyond this, this study encompasses a comprehensive investigation into the multifaceted dynamics involved in dynamic pricing within microgrids, the strategic methodologies of demand response, and the burgeoning energy demands prevalent in high-consumption regions.

The complexity of these dynamics has sparked a need for a diverse exploration of optimization models and algorithms. A primary contribution of this research effort involves the conceptualization and implementation of an intricately designed, energy-efficient scheduling algorithm. This algorithm was meticulously crafted with the explicit goal of optimizing residential appliance usage, reducing energy expenditure, and alleviating the

strain on energy suppliers, thereby fostering a more sustainable and economically viable energy consumption paradigm.

Our research methodology is inclusive of essential components such as comprehensive data collection, intricate predictive modeling techniques, algorithm design tailored for efficiency, rigorous performance evaluation, and in-depth comparative analysis. Through the synthesis of these critical components, we aim to provide a thorough and holistic understanding of the interwoven complexities within energy management systems.

2. Literature Review

The evolution of real-time scheduling systems has led to the development of fixed-priority scheduling theories, facilitating task prioritization based on start times, computation times, and deadlines [1]. In the realm of scheduling problems, various approaches, including machine learning and artificial intelligence, are employed, necessitating trade-offs between solution quality and computational time [2]. A critical challenge arises in the form of the resource-constrained project scheduling problem (RCPSP), involving the management of activities with limited resources and capacities [3]. Researchers have explored a multitude of methods, encompassing both heuristic and mathematical techniques, primarily focusing on time constraints [4–6]. The emergence of the multi-mode RCPSP introduced multi-mode resource constraints, with solutions rooted in both mathematical and heuristic methodologies [1,7–9].

In the realm of dynamic pricing in microgrids, challenges related to real-time information availability and customer scheduling are encountered. The use of reinforcement learning dynamic pricing algorithms has been proposed as a cost-reduction strategy, albeit sometimes overlooking user satisfaction [10]. Demand response relies on real-time energy price information, empowering customers to schedule devices based on smart grid data [11,12]. Diverse pricing plans, including time-of-use pricing and reinforcement learning algorithms for handling unpredictability, have been explored [13–15]. Different models consider the maximization of customer utility or supplier profit, often through dynamic pricing policies [16]. The growing demand for electricity, particularly in high-energy-consumption areas, has exacerbated supply–demand disparities. The introduction of microgrids and residential energy management is intended to curtail power consumption by implementing energy-saving and renewable sources [17,18]. Demand response algorithms prioritize matching power demand to reduce costs and consumption, frequently utilizing time-based rates or financial incentives for customers. A common strategy involves using linear programming models to optimize demand and response, especially in microgrids, utilizing mixed integer approaches and constraints to schedule power consumption across various appliances and time slots [19]. To enhance solution quality and accuracy, optimization techniques such as particle swarm optimization and stochastic methods are employed [3].

A range of optimization models have been introduced, encompassing diverse strategies like the Gray Wolf Optimization algorithm, which necessitates energy management controllers and smart meters [1]. Additionally, alternative approaches, including the greedy algorithm, mapping algorithms, and stochastic dynamic optimization, have been explored [20]. Some models focus on improving the quality of the user experience and minimizing power consumption [21]. Linear optimization has also been employed from the microgrid manager's perspective, aiming to maximize profit by minimizing costs [22]. Stochastic dynamic optimization is appropriate for complex constraints, considering multiple targets to minimize costs [23]. Moreover, energy storage and optimization are employed to mitigate peak demand [24,25]. Furthermore, linear optimization is applied to optimize peak loads and real-time scheduling algorithms [26], collectively enhancing energy management and reducing peak power consumption.

Optimizing microgrid energy usage involves various objectives, including profit maximization, cost reduction, and renewable energy utilization [23]. The choice of algorithms and model components varies between studies, with some exploring scenarios involving

energy storage technologies [27]. Quality of User Experience (QoE) constraints are considered, assessing the impact of load scheduling on consumers [21,27]. Notably, substantial cost savings (ranging from 19% to 84%) have been demonstrated by optimizing renewable production and consumption using QoE-aware algorithms [27]. Another innovative approach involves modeling the use of plug-in electric vehicles as an energy storage solution for households [28]. Game theory algorithms are applied in certain studies to maximize microgrid profits while minimizing costs [29–31]. Some studies incorporate technical constraints such as thermal limits and bus voltages to manage energy within and between microgrids [31]. The integration of distributed generation and the consideration of price elasticity have been employed to reduce system losses and enhance stability [31]. The application of the multiple-leader-multiple-follower Stackelberg game, with microgrids acting as leaders and customers serving as followers, has addressed storage availability and optimum pricing [32].

The context-aware energy management paradigm, often leveraging cloud technologies, optimizes data derived from sensors and devices within smart home energy management systems. The data are harnessed for learning user behaviors and predicting energy requirements [5]. Notably, Kazemi et al. [1] introduced a cloud-based framework designed to conserve electricity and reduce carbon dioxide emissions, relying on data from sensors and user behavior. The Context-Aware Energy Management System (CAEMS) introduced by [5,14] leverages data-mining algorithms to predict context information, facilitating the proactive control of energy resources and user preferences. In addition, Ruelens et al. [15] developed an energy management model using a Hierarchical Hidden Markov Model, drawing on activity patterns and real-time data. Location prediction, highlighted by Wang et al. [33], has proven instrumental in minimizing energy consumption while enhancing user comfort. The Internet of Things (IoT) plays a pivotal role in enhancing energy efficiency in public facilities, as demonstrated by IMPReSS [34]. Moreover, the studies by Yaagoubi et al. [35] and Al-Mulali et al. [36] have proposed user-comfort-aware load management algorithms, striking a balance between energy costs and user comfort. Pedrasa et al. [37] employed particle swarm optimization, while Mohsenianrad et al. [16] conducted an analysis of energy consumption scheduling within a dynamic pricing environment. Sinaki et al. [38] utilized a knapsack-based method for appliance scheduling, albeit they faced challenges related to prioritization. A spectrum of approaches, including mixed techniques and mathematical models, were proposed by Jain et al. [39] and Grosse et al. [40] for direct load control and demand-side management during peak hours. Shi et al. [41] proposed an equation for modeling home appliances while addressing electricity costs.

The evolution of energy consumption optimization in dynamic pricing environments has been substantially propelled by a spectrum of scholarly inquiries employing machine learning methodologies. Similarly, the integration of the Cognitive IoT for monitoring energy consumption patterns [42] has showcased commendable accuracy in predictive analytics for electricity bills and anomalous energy usage detection. Furthermore, investigations into time series forecasting for stock prices [43] and household appliance energy consumption prediction [44] have revealed robust performance, particularly in complex and ever-changing settings. Moreover, the amalgamation of supervised and transfer learning strategies [45] has led to notable efficiency gains in optimizing energy systems, while the advent of TinyML within IoT frameworks [46,47] introduced multiple pathways with which to confront environmental challenges via the integration of smart ML-enabled embedded systems that may be used to make intelligent decisions and control other devices such as the home appliances in our context. Additionally, the adeptness of deep Q-learning and double deep Q-learning in managing residential energy consumption [48] underscores their effectiveness in adaptively reducing electricity usage in fluctuating environments.

This literature review uncovers a notable research gap in effectively applying and validating theoretical models within the practical, complex, and variable contexts of real-world settings. This gap is particularly evident in the field of energy management, where current models primarily focus on cost reduction and energy consumption optimization,

frequently at the expense of user comfort. Such an oversight underscores an urgent need for research that not only fine-tunes energy use but also earnestly incorporates user comfort and satisfaction. This balanced approach is vital in ensuring that energy management strategies are both efficient and user-friendly in practical scenarios.

Addressing this issue, in our study, we set out to develop a user-centric, adaptive model that bridges the divide between theoretical constructs and their tangible application in dynamic real-world environments. Our model is specifically designed to strike a harmonious balance between energy efficiency and user comfort, ensuring its relevance and applicability in various real-life situations. By focusing on this equilibrium, we aim to showcase that achieving sustainable energy savings is feasible without compromising on the quality of user experience. This endeavor aligns with our broader objective of proposing a model that is not only theoretically sound but also practically viable and effective in enhancing both energy management and user satisfaction in a dynamic pricing environment.

3. Methodology

In this section, we delineate the methodological framework underpinning our research endeavor. Our study revolves around the development and evaluation of an energy-efficient scheduling algorithm for optimizing appliance usage within residential environments. The primary objective is to minimize energy costs and alleviate the load on energy suppliers, fostering sustainable and economic energy consumption. The implementation of this methodology encompasses the following key components: data collection, predictive modeling, algorithm design, performance evaluation, and comparative analysis.

3.1. Data Collection and Preprocessing

The core of our research methodology hinges on the acquisition of a comprehensive dataset sourced from the Transmission Service Operator (TSO) Red Electric Espana via the European Network of Transmission System Operators for Electricity public portal. Spanning a four-year window, this dataset is a valuable repository of insights into Spain's electrical generation and pricing dynamics [49]. Our data augmentation strategy incorporated supplementary sources, encompassing weather data and holiday information from five major Spanish cities: Madrid, Barcelona, Seville, Granada, and Palma de Mallorca. This augmentation process incorporates meteorological parameters, such as temperature and humidity, recorded at hourly intervals, as well as binary indicators for holiday occurrences and one-hot-encoded representations of weekdays [50,51].

The inclusion of weather and holiday data is pivotal, given their substantial influence on energy consumption patterns. Additionally, our data analysis entails the detection of week-based seasonality, a critical factor for the accurate modeling of energy prices. Figure 1 illustrates the clear week-based cycles in energy consumption [52].

Data preprocessing constitutes a crucial phase following data acquisition, with the primary objective of ensuring the integrity of data and optimizing their utility for model development. Several key preprocessing steps were carried out:

1. **Normalization:** This step ensures data uniformity by scaling electricity prices and loading demand data to a common reference point, typically the mean or median. The normalization process promotes equitable contributions of features to model training and prediction, irrespective of their scale or magnitude [53].
2. **Feature Engineering:** Incorporating supplementary data necessitates thoughtful feature engineering. Categorical variables, like weekdays, are one-hot-encoded to facilitate their integration into predictive models. Feature engineering also encompasses the creation and transformation of variables to capture nuanced data relationships. Lagged features and moving averages are computed to capture temporal dependencies and trends in the dataset [23].
3. **Data Partitioning:** Data partitioning is a fundamental step in facilitating model training and evaluation. The dataset is divided into training, validation, and test sets. The training set is dedicated to model training, the validation set assists in hyper

parameter tuning, and the test set provides an unbiased evaluation of model performance. To preserve the temporal order of data, time-series cross-validation is often employed [54].

4. **Outlier Handling:** Robust statistical methods and domain-specific knowledge guide the detection and management of outliers in the dataset. Outliers, stemming from irregular events or measurement errors, are addressed to prevent undue influence on model training and predictions [55].
5. **Missing Data Handling:** Given the real-world nature of the data, values may be missing. To address this, missing data are imputed using strategies such as linear interpolation to maintain data completeness and mitigate data quality issues [56].

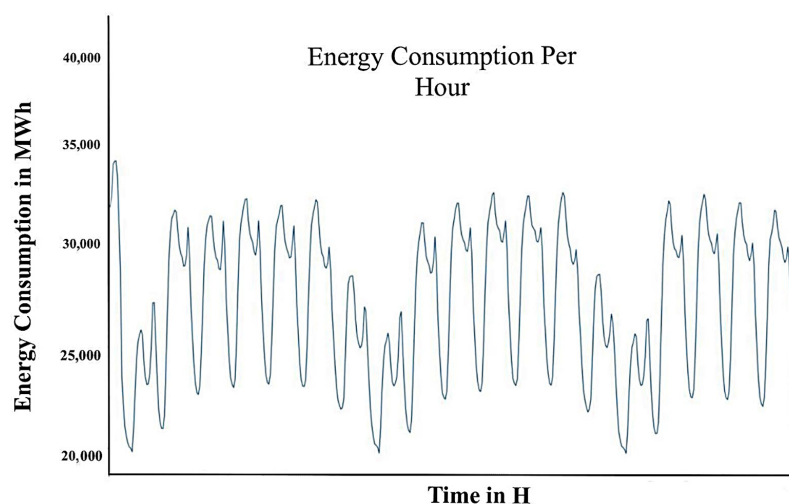


Figure 1. Graph of week-based cycles of energy consumption [52].

3.2. Algorithm Design

The algorithm's initiation process involves obtaining the user's appliance selection. Subsequently, the algorithm categorizes the appliances into three distinct classes, each necessitating a tailored approach:

1. **Fixed-Load Appliances:** This category encompasses devices that must be utilized when the user requires them, without any temporal adjustments. For fixed-load appliances such as home lighting and computers, immediate operation is essential. In such cases, the algorithm promptly recommends the current date and time as the optimal choice, given the expectation of immediate device activation. A minimum heap is employed to extract the time when energy expenses are at their lowest, disregarding time contiguity. The algorithm returns N smallest numbers from the predicted list.
2. **Regulatable Load:** In the case of regulatable loads, which include appliances like water pumps, users have the flexibility to regulate these devices to achieve peak efficiency. Unlike fixed-load appliances, regulatable loads do not require immediate operation. The algorithm considers the due date, ensuring that the required output is achieved within the specified timeframe, thereby guaranteeing user satisfaction.
3. **Deferrable Loads:** Devices that are characterized by deferrable loads such as washing machines possess the capacity for delayed operation but necessitate continuous operation within a contiguous time frame for optimal performance.

The main algorithm employs the Kadane Algorithm, a dynamic programming technique, to identify the contiguous time slot within the energy price predictions dataset where energy costs are minimized, all while accommodating the appliance's average usage duration. This optimization process is achieved through a sliding window approach, characterized by the following steps. First, a dynamic window with a width equal to the appliance's average usage time is systematically shifted across the dataset of energy price

predictions. Second, during each iterative step, the algorithm computes the cumulative energy price within the window. Third, upon traversing the entire dataset, the algorithm identifies the group of contiguous time slots that collectively yield the minimum energy cost, thus signifying the optimal window for device operation. Within this framework, the `minSubsequent()` and `Kmin-Heap()` functions return a date and time value, representing the most advantageous moment for appliance operation. In the context of fixed-load appliances, the due date can be defined as the current date and time, accounting for the fact that these devices necessitate immediate activation.

The scheduling algorithm is designed to empower users with the ability to make well-informed decisions regarding the timing of appliance usage guided by predictive energy-pricing data. It effectively balances user preferences and cost-efficiency considerations, thereby enhancing overall user satisfaction. Figure 2 furnishes a comprehensive flowchart illustrating the intricate decision-making process and logical steps inherent in our scheduling algorithm's operation.

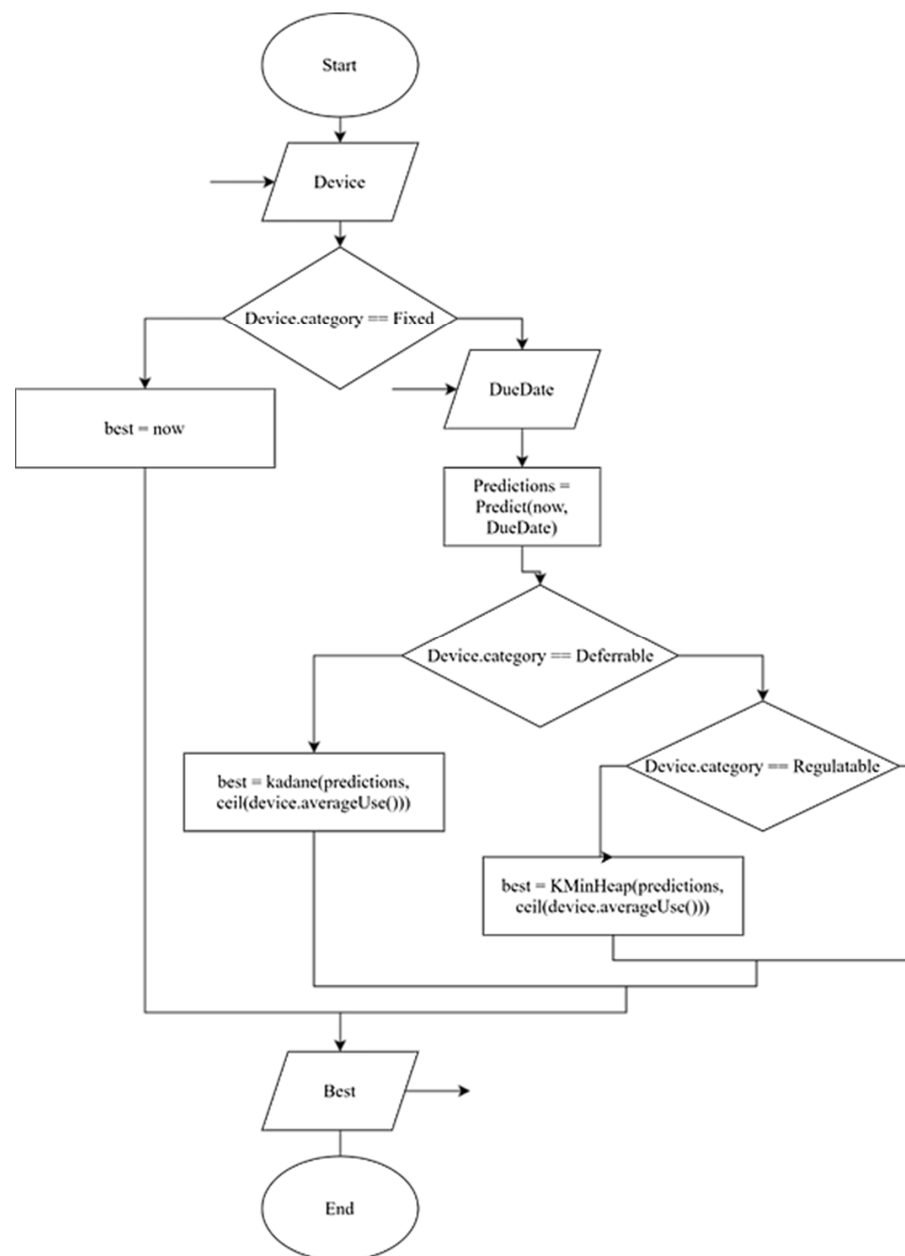


Figure 2. Flowchart of our proposed algorithm.

3.3. Scheduling Algorithm—Performance Metric

Over the last decade, deep learning has emerged as a prominent and increasingly utilized methodology in academic circles. It operates through deep networks incorporating specialized and more adept unit neurons. Recurrent Neural Networks (RNNs) stand out as a unique neural unit designed to retain and amalgamate information from preceding events [57]. However, in the application of RNNs, one encounters issues related to gradients, such as vanishing or exploding gradients, which hinder effective weight updates [27]. These challenges often result in insufficient learning on the part of the network.

To circumvent these issues and harness the persistence feature effectively, specialized techniques such as Long Short-Term Memory Networks (LSTMs) and Gated Recurrent Units (GRUs) have proven significantly effective. LSTMs have demonstrated their superiority over traditional neural networks and are unique in that they do not suffer from the vanishing gradient problem [58]. They manage and modify memory states by using an activation function to forget specific parameters across the neuron's path, simulating the process of forgetting. LSTMs effectively avoid gradient issues by utilizing memory cells and gates to retain gradients within the cell. Figure 3 illustrates the architecture of the LSTM network adopted in our algorithm.

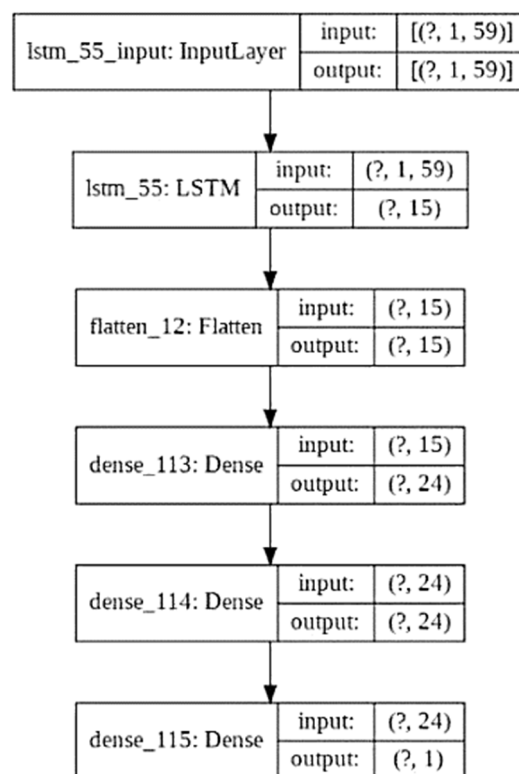


Figure 3. Summary of the used LSTM model.

Similarly, GRUs, initially conceived as a streamlined version of LSTMs, excel in managing long-term dependencies and evading the issues associated with vanishing gradients through the application of forget units, thereby optimizing time lags in a model.

From a temporal complexity perspective, the analysis of artificial neural networks involves two primary processes: back propagation for updating network weights and biases and forward propagation for generating predictions [59]. Back propagation occurs during the training phase of predictive artificial-neural-network-based models, whereas the forward pass is pivotal for real-time prediction generation during system utilization. Computational complexity predominantly lies in the training process, rendering the forward pass computationally cost-effective and amenable to less-sophisticated hardware.

A comparative evaluation of neural-network-based models and multi-linear regression techniques reveals a pronounced degree of superiority in the performance of neural networks. Within this category of models, the Long Short-Term Memory (LSTM) network demonstrates the most commendable performance, registering a Mean Absolute Error (MAE) of 5.32. This is closely followed by the Gated Recurrent Unit (GRU), which exhibits an MAE of 7.41. In contrast, a standard implementation of a neural network, often referred to as a ‘vanilla’ neural network, records a relatively higher MAE of 8.0. These findings, as delineated in Table 1, clearly highlight the hierarchical performance distinctions among the various neural network architectures in comparison to multi-linear regression methods. Figure 4 presents a comprehensive graphical representation that juxtaposes the predictive outputs of the Transmission Service Operator’s (TSO) model with those derived from our proposed LSTM-based model, set against the backdrop of the actual energy prices [59].

Table 1. Performance comparison of various predictive models.

Model	Description	MAE	MSE	RMSE
Neural Network	3 layers containing neuron quantities of 20, 12, and 1, respectively	8.05	92.89	9.63
Multi-Linear Regression	Inapplicable	9.57	130.3	11.41
GRU	15 GRU units, a flattening laying, and 3 layers containing neuron quantities of 15, 8, and 1, respectively	7.41	80.49	9.32
LSTM	15 LSTM units, a flattening laying, and 3 layers containing neuron quantities of 24, 8, and 1, respectively	5.32	48.90	6.99
TSO Model	Inapplicable	9.68	131.1	11.48

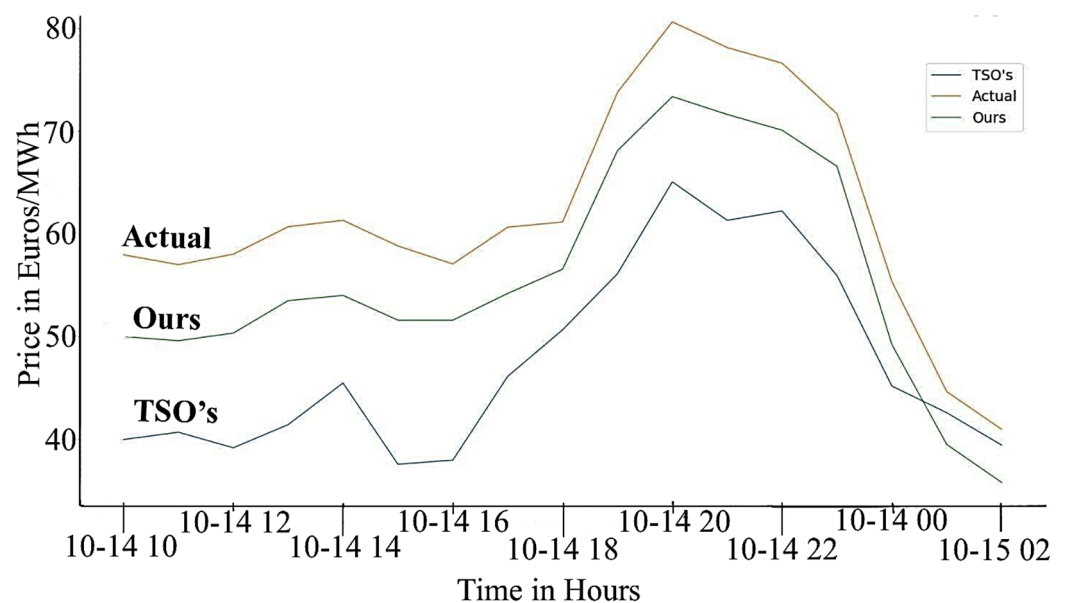


Figure 4. Comparison of actual energy prices, TSO’s predictions, and the predictions made by our LSTM-based model [59].

The dataset employed in forecasting energy prices offers valuable insights into the performance metrics of diverse predictive models. A notable parallelism can be observed

between the multi-linear regression approach and that of a conventional model, identified as the transmission service operator (TSO), which exhibits MAE errors of 9.57 and 9.68, respectively. This similarity in performance metrics potentially suggests underlying commonalities in the methodologies employed by these two predictive approaches.

To explicate the marked differences in performance between the models, the consideration of a neural network devoid of activation functions and hidden layers, resulting in a collection of an activationless perceptron akin to a linear regression model, is instructive. This simplification helps to highlight the fundamental differences between linear and non-linear models. The expansive nature of neural networks and their superior adaptability in handling nonlinear fittings elucidates their efficacy in surpassing or at least equaling the performance of linear regression models. Neural networks provide a more nuanced and sophisticated approach to prediction tasks by leveraging their capacity to learn intricate, nonlinear relationships present in a dataset.

In the realm of LSTM networks, an advantage over traditional artificial neural networks can be observed. LSTM networks amalgamate the foundational attributes of their predecessors while concurrently addressing and rectifying prevalent challenges. This rectification is primarily facilitated through the distinctive memory cell structure inherent in LSTMs. This structure is adept at efficiently encapsulating long-term data dependencies, a capability that is particularly instrumental in time-series analysis. The efficacy of LSTM networks in this regard stems from their ability to integrate and interpret historical data contexts, thereby yielding more accurate predictive outcomes in scenarios where temporal continuity and past data relevance are critical.

3.4. Predictive Modeling

In this section, rather than solely assessing the predictive model's performance, the focus is on employing the root-mean-squared error (RMSE) to gauge the accuracy of a scheduling algorithm. The goal is to define the optimal time objectively by using actual consumption data from the dataset rather than relying solely on the model's predictions. This involves selecting the actual minimum energy prices, which, in turn, influence the RMSE equation given as follows (1):

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (Y_i - \hat{Y}_I)^2}{n}} \quad (1)$$

The RMSE method penalizes significant deviations between predicted and actual values, emphasizing errors without regard to their sign. By quantifying the average squared error and taking its square root, this metric provides a measure of the difference between the real and predicted value vectors [60].

The performance evaluation outlined in Table 2 demonstrates the superiority of the scheduling algorithm over the current standard practice. This algorithm exhibits near-optimal performance as the time offset decreases. This improvement is due to increased possibilities for optimal actions as the offset window diminishes. Consequently, similar forecasting-based methods exhibit poorer performance with larger offsets. This suggests that without accounting for changes in energy prices over time, identifying periods of minimal energy costs becomes challenging. The prevalent behavior that reproduces the most common patterns contributes to observed peaks in energy demand.

In real-life scenarios, this algorithm's predictive accuracy diminishes as the time difference from the present increases. However, setting a threshold at around 12 h in real-world applications ensures this algorithm's reliable performance and mitigates potential underperformance caused by extended offsets.

Table 2. Results regarding root mean square error for the scheduling algorithm.

Offset	Schedules	RMSE	
		With Scheduling (Paper Approach)	Without Scheduling
2 h	Light	0.23	2
	Medium	0.26	1.98
	Heavy	0.32	1.98
6 h	Light	1.16	4.48
	Medium	1.36	4.78
	Heavy	2.89	5.89
12 h	Light	2.56	10.85
	Medium	3.12	11.98
	Heavy	3.89	11.98

4. Results

To evaluate our approach's effectiveness, reliability, and accuracy, we employed a simulated environment. This environment allowed us to adjust parameters efficiently and maintain a level of numerical accuracy similar to real-life scenarios while reducing replication time. The central element of our system is an entity tasked with time management, device definition, scheduling appliance operations, and the creation of statistical representations.

Our system automatically generated schedules based on weekly lifestyle data, enabling the creation of realistic, extended schedules from shorter seven-day calendars. Table 3 details the appliances considered during our testing. The World object served as the simulated environment for testing appliances and schedules, mimicking real-world behavior. While the time object synchronized simulation time with real-world data, fixed-load appliances were not extensively modified due to their significant impact on user satisfaction. These appliances are expected to operate without delay, ensuring, for example, immediate TV viewing upon activation.

Table 3. Appliances used in the simulation and their respective amounts of energy consumption.

Appliance	Type	Consumption
Washing Machine (M)	Deferrable	1000 W/h
Dish Washer (D)	Deferrable	1800 W/h
Oven (O)	Deferrable	2300 W/h
Electric Vehicle (V)	Regulatable	5000 W/h
Water Pump (P)	Regulatable	800 W/h
Water Heater (H)	Regulatable	1300 W/h
Home Lights (L)	Fixed	200 W/h
TV (T)	Fixed	300 W/h

This project integrates an automated simulation capability with interactive hour-by-hour manual settings, providing users with the ability to actively engage with the system, mimicking real-life interactions for easy implementation. In the setup function, the declaration of appliances involves specific parameters: Appliance Categories, which are crucial for distinct operations based on the appliance category; optional Appliance Labels, enhancing user understanding through appliance identification; optional Appliance Average Runtime, which assists the system in determining a suitable window size for the

minimum value and is automatically updated during device use; and Appliance Average Consumption, which is finely tuned for generating comparative results and accurately representing system performance.

The user is initially prompted to schedule appliances, selecting the device, desired activation date, and scheduling algorithm. The system predicts electricity prices and schedules device operation during the most cost-efficient periods. Users can view various statistics, such as average usage time, number of uses, total and historical consumption, and expenses for each declared appliance.

Control over the simulated world's time offers two options: users can increment time by an hour or utilize real-time synchronization. The real-time synchronization prompts users to set the scale for time synchronization. For instance, selecting a minute means that the world's time will advance each minute in real time. To implement this project in real-life scenarios, setting the synchronization scale to match hours ensures the system synchronizes in real time, updating simultaneously at equal rates.

In this section, to better understand the performance of our algorithm, we introduce abbreviations for various appliances (shown in Table 3). The schedules, represented by Figures 5–7, correspond to light, medium, and heavy appliance usage, respectively. This differentiation allows for a comprehensive evaluation of the algorithm under diverse real-life conditions. The light schedule represents minimal appliance usage, suitable for smaller households, encompassing basic appliances like lights and occasional kitchen appliances. The medium schedule mirrors average household appliance usage, including regular use of water heaters, washing machines, and kitchen appliances. The heavy schedule caters to larger families with higher energy demands, featuring the extensive use of all major household appliances. By simulating these scenarios, the research can demonstrate the flexibility and effectiveness of the scheduling algorithm in real-world settings, showcasing its potential to reduce energy costs and optimize usage across a spectrum of household demands. This approach is essential in validating the algorithm's practical applicability and effectiveness in diverse living environments.

TD	M	T	W	H	F	S	U
10				M			
11	O		O		O		
12							
13							
14							
15	M						
16							
17							
18							
19							
20							
21	D	D	D	D	D	D	D
22							

Figure 5. Light schedule.

TD	M	T	W	H	F	S	U
10	H	H	H	HM	H	H	H
11	O		O		O		
12							
13					L		
14		L	L		L		L
15	M	L					L
16							L
17			L		L		
18			L				
19							
20	LP	LP	LP	LP	LP	LP	LP
21	LTD	LTD	LTD	LTD	LTD	LTD	LTD
22	LT	LT	LT	LT	LT	LT	LT
23	LT	LT	LT	LT	LT	LT	LT

Figure 6. Medium schedule.

TD	M	T	W	H	F	S	U
10	HP	HP	HP	HP	HP	HP	HP
11	OP	OP	OP	OP	OP	OP	OP
12	T	O	TL	O	TL	O	TL
13	T	TL	TL	T	TL	T	TL
14		TL		T		T	
15	ML	LT	ML	LT	ML	LT	ML
16	L	LT	LT	OLT	LT	LT	L
17	HL	HL	HL	OHL	HL	HL	HL
18	L	L	L	L	L	L	L
19	L	L	L	L	L	L	L
20	LP	LP	LP	LP	LP	LP	LP
21	LTD	LTD	LTD	LTD	LTD	LTD	LTD
22	LT	LT	LT	LT	LT	LT	LT
23	LT	LT	LT	LT	LT	LT	LT

Figure 7. Heavy schedule.

Figure 5 outlines a basic schedule, serving as a starting point for our algorithm's evaluation. It features three primary devices: an oven, employed three times a week during lunch hours; a washing machine, used twice a week; and a dishwasher, operating five times weekly. This stripped-down setup allowed us to gauge the initial performance and efficiency of our algorithm on a small scale.

Moving to Figure 6, we observe a more moderate household routine, providing a broader spectrum of appliance usage:

- The water heater is operated daily at 9 a.m.;
- The washing machine is run twice a week, scheduled for 2 p.m. and 10 p.m.;
- The oven sees use four times a week, primarily during lunch hours on Mondays, Thursdays, and Sundays, with an additional session on Wednesdays in the afternoon;
- The water pump is utilized daily in the evening;
- The dishwasher aligns its usage with the oven's schedule, running five times a week;
- Lights are scheduled daily, illuminating the household from 7 p.m. until 11 p.m.

Finally, Figure 7 showcases a more extensive appliance utilization scenario, providing insight into our program's effectiveness in a realistic environment:

- The water heater is employed twice a day, namely, at 9 a.m. and 5 p.m.;
- The washing machine is operated every other day, specifically at 2 p.m.;
- The oven is in use daily at 11 a.m.;
- The water pump is utilized twice a day—at 9 a.m. and 7 p.m.;
- The dishwasher operates daily at 9 p.m.;
- Lights are switched on from 2 p.m. to 4 p.m. and again from 7 p.m. until 11 p.m. daily.

Table 4 presents a comprehensive summary derived from executing the simulation using the medium schedule alongside a 6 h offset period over six months. It offers a comparative analysis of energy costs from January to June 2018, specifically utilizing the medium schedule outlined in Figure 6. Notably, when no scheduling algorithm is employed, the energy costs witness a significant reduction when contrasted with the prevailing conventional practices. However, concerning scenarios involving minimal devices and runtimes, such as in the case of lighting, the algorithm's optimization capabilities are constrained. Conversely, in the context of a heavier usage schedule, a substantial price decrease of 38.85% occurs, indicating a considerable reduction in energy expenses compared to scenarios with no schedule. It becomes evident that the energy consumption costs are notably lower across all the algorithms implemented, with a direct correlation between the reduction in costs and increased appliance usage. The degree of bill reduction is directly proportional to energy usage, as per the user's schedule.

Table 4. Summary of energy cost reductions achieved using various prediction algorithms on increasingly loaded schedules over 6 months.

Schedules	% Cost Reduction				
	MLR	NN	GRU	TSO	LSTM
Light	7.86	10.01	10.38	7.77	11.11
Medium	11.21	13.89	18.26	11.07	20.09
Heavy	30.87	31.58	34.42	30.15	38.85

Moreover, an influential factor impacting the proposed approach's performance, compared to that in scenarios lacking scheduling, is the timing of device usage. Closer proximity to peak hours and the temporal distance from the current date and time significantly influence reductions in energy bills and overall load demand. Notably, cost reduction exhibits substantial escalation from one schedule to another, with the proposed approach demonstrating more efficient performance with specific devices, notably electric vehicles. The algorithm's ability to markedly reduce energy costs is particularly evident for cases like electric vehicles, which offer charging periods as short as 30 min, allowing the algorithm to

optimize energy procurement to a significantly greater degree than could be achieved via regular human precision.

5. Conclusions

This study comprehensively explores the multifaceted domain of energy scheduling and optimization within microgrid and residential environments. It addresses the evolution of real-time scheduling systems, challenges in dynamic pricing, and the array of optimization strategies applied. Through methodological insights, this paper showcases the intricacies involved in designing energy-efficient scheduling algorithms that minimize costs and optimize appliance usage.

The findings illustrate the remarkable efficiency of the proposed scheduling algorithm, which uses LSTM-powered artificial neural networks. This algorithm outperforms traditional models across various metrics, exhibiting a notably lower Mean Absolute Error (MAE) of 5.32 compared to that of Neural Networks (8.05) and Multi-Linear Regression (9.57). This heightened predictive accuracy directly correlates with improved energy efficiency and cost savings. This research emphasizes the significance of balancing user satisfaction, cost reduction, and effective scheduling for sustainable energy consumption. The study demonstrates a substantial 38% improvement in optimized schedules, underscoring the algorithm's practical impact on energy efficiency and substantial cost reduction. Overall, this work presents invaluable insights and methodologies pivotal in advancing energy management and addressing complexities in modern energy-scheduling systems.

The efficacy of this study is closely linked to the integrity and extent of its data sources, with the performance of the model being particularly reliant on the quality and comprehensiveness of the available data. This dependency on data highlights a need for future research to address scenarios characterized by limited or incomplete datasets. Additionally, the complexity of the proposed scheduling algorithms, especially those employing advanced neural network architectures with large quantities of parameters, poses significant challenges in terms of computational demands and scalability, especially in environments with limited resources, such as low-energy machine-learning-enabled microcontrollers (TinyML). The issue of generalizability also arises, as the algorithms and methodologies developed may be highly specific to the datasets and scenarios used in this study. Their applicability across different geographical regions, varying microgrid configurations, or diverse consumer behavior patterns is an area that warrants further exploration.

Future research avenues appear promising, particularly in developing adaptive models that can perform robustly with varying levels of data quality, ensuring effective performance even in the presence of limited or noisy datasets. The scalability of the proposed models and their adaptability to different types of microgrids and consumer environments represent a key area for future exploration. The integration of real-time data, such as immediate weather changes or unplanned power outages, into the scheduling algorithm is another potential enhancement that could significantly increase its accuracy and practical utility. Delving deeper into user-centric approaches, future research could explore more sophisticated methods to quantify and incorporate user comfort and preferences into the scheduling algorithms. The role of energy storage solutions and the integration of renewable energy sources in the scheduling algorithms is also a crucial area for future investigation. Improving the transparency and interpretability of complex algorithms like LSTMs could enhance their acceptability and trustworthiness among users. Collaborations with other fields, such as behavioral science and urban planning, could provide new insights into energy consumption patterns and aid in developing more holistic energy management solutions. Finally, understanding and influencing policy and regulatory frameworks to support the implementation of advanced scheduling algorithms in energy management together represent another significant direction for future research.

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